Odd Mid Semester Examination, September-2025

Name of the Program: B.Tech. Biotechnology — *Semester: III*

Name of the Course: Genetics and Molecular Biology — *Course Code:* **TBT-301**

Time: 1-1/2 Hour — Maximum Marks: 50

Note:

(i) Answer **all the questions** by choosing ***any one of the sub questions***.

(ii) Each question carries 10 marks

Q1	(10 marks)	CO1
(a)	Describe Mendel's monohybrid and dihybrid crosses. How do they demonstrate the principle of segregation and independent assortment?	
OR		
(b)	Discuss quantitative inheritance (two gene interaction) with a suitable example of inheritance of skin color of human.	
Q2	(10 marks)	CO1
(a)	State the postulates of Gene theory of Heredity and corelate with Mendel's Law.	
OR		
(b)	What is the basis of Chromosomal Mapping? What would be the distance of two genes on chromosome if 10 recombinants were produced out of 100 offsprings?	
Q3	(10 marks)	CO1
(a)	What is population genetics? Calculate the frequency of allele 'M' and 'm' in a Mendelian population, if 120 individuals are of genotype 'MM', 10 individuals are of genotype 'mm', and 70 individuals are with genotype 'Mm'.	
OR		
(b)	What are sex-linked traits? Explain their inheritance in humans with the example of hemophilia.	
Q4	(10 marks)	CO2
(a)	With the help of a well-labeled diagram, analyze and explain the salient structural features of Watson and Crick's Double Helix Model of DNA.	
OR		
(b)	Write short notes on any two- 1). C-value paradox; 2). Cri-du-chat syndrome; 3). Klinefelter syndrome; 4). CoT curve	
Q5	(10 marks)	CO2
(a)	Differentiate between the roles of covalent bonds, hydrogen bonds, and hydrophobic interactions in maintaining DNA stability.	
OR		
(b)	Describe the process of DNA denaturation and renaturation. How is melting temperature (Tm) determined?	